Supplementary Material

What Near-Perfect Accuracy Does and Does Not Show on the UCI TCGA Pan-Cancer RNA-Seq Dataset: Leakage, Equivalence, Conformal Reliability and Data Efficiency

Contents: S1 Software, settings and data integrity; S2 Per-class performance and pooled errors; S3 Feature-count ablation; S4 All-gene benchmark; S5 All pairwise comparisons; S6 Leakage audit; S7 Misclassified samples and quality metrics; S8 Top-ranked genes and selection stability; S9 Computational footprint and fold-level distributions; S10 Reproducing the results (Supplementary File S1); S11 Calibration and conformal prediction (robustness checks, all models); S12 Learning curves and leakage by sample size, with 95% CIs.

S1 Software, settings and data integrity

All analyses were run with Python 3.13.5 on a 8-vCPU Linux virtual machine (x86-64), using scikit-learn 1.9.1, LightGBM 4.7.0, NumPy 2.5.4, pandas 3.0.6, SciPy 1.18.1 and Matplotlib 3.11.2 and joblib 1.6.0 (pinned in requirements.txt). The main benchmark took 18 min; the revision analyses (leakage audit, all-gene benchmark) run in parallel and add about 1 h on 8 vCPUs, mostly LightGBM on all genes; the extension and robustness analyses add about 5 min. The dataset was downloaded from the UCI repository (dataset 401) on 16 October 2026; SHA-256 checksums: data.csv e3e7eff51f4a5fe5f121fb183c09786c767c755ef10d1c7b3ce3c193a7082ca9, labels.csv e1653ccf5a630deb3b953ccac5a8b331b89adc5433e5d2d3444c1fc01d79e7aa.

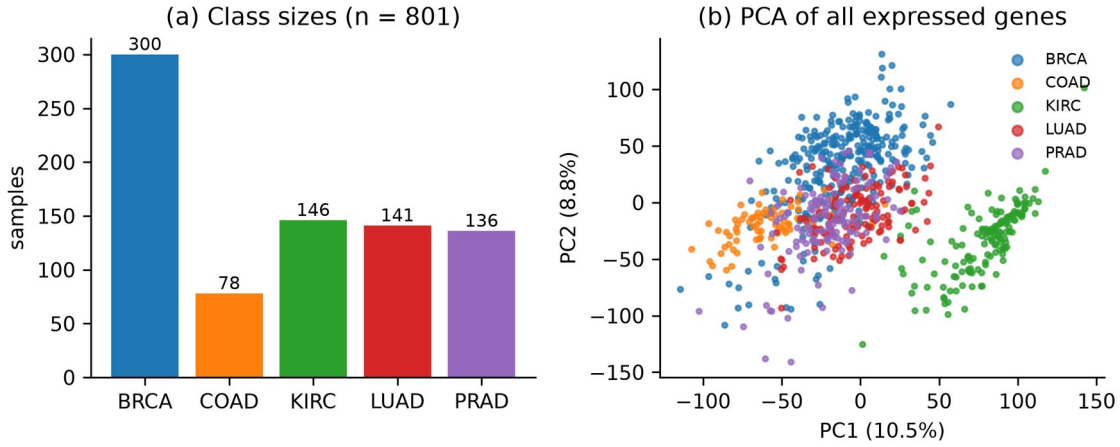


Figure S1. Dataset overview. (a) Samples per cancer type. (b) First two principal components of the standardised expression of all non-constant genes, fitted on all 801 samples for visualisation only.

Table S1. Fixed settings (no hyperparameter tuning)

Component	Setting
Majority baseline	most frequent class (BRCA)
Softmax (multinomial LR)	L2 penalty, C = 1, lbfgs, max_iter 5000
Ridge classifier	α = 10; one-vs-rest least squares
Linear SVM	LinearSVC, C = 1, squared hinge, one-vs-rest, max_iter 20000
Gaussian naive Bayes	default variance smoothing 10^{-9}
Random forest	500 trees, Gini, √p features per split, seed 2026
LightGBM	200 trees, learning rate 0.05, 15 leaves, seed 2026
MLP	one hidden layer, 64 ReLU units, Adam, L2 10^{-3}, max 1000 epochs, seed 2026
Pipeline (all)	zero-variance filter → ANOVA-F top-k (k = 100) → z-scoring → classifier; refitted per fold
Cross-validation	RepeatedStratifiedKFold, 5 folds × 10 repeats, seed 2026; identical splits for all models

S2 Per-class performance

Table S2. Per-class F1 and balanced accuracy

Model	BRCA	COAD	KIRC	LUAD	PRAD	Balanced acc., % [95% CI]
Softmax (multinomial LR)	0.994	0.995	0.998	0.988	1.000	99.32 [98.48, 100.00]
Ridge classifier	0.996	1.000	1.000	0.991	1.000	99.63 [99.07, 100.00]
Linear SVM	0.998	0.998	1.000	0.995	1.000	99.75 [99.21, 100.00]
Gaussian naive Bayes	0.993	0.987	0.993	0.972	1.000	98.93 [98.14, 99.73]
Random forest	0.996	0.994	0.998	0.989	1.000	99.38 [98.59, 100.00]
LightGBM	0.995	0.993	0.999	0.987	0.996	99.32 [98.53, 100.00]
MLP (64 hidden units)	0.995	0.993	0.997	0.988	1.000	99.28 [98.41, 100.00]

Per-class F1 from the confusion matrix summed over 10×801 out-of-fold predictions; balanced accuracy: mean over 50 folds with corrected 95% CI. k = 100.

Table S3. Pooled error rates

Model	Errors	Error rate, % [Wilson 95% CI]	Perfect folds, %
Majority baseline	5010	62.55 [61.48, 63.60]	0
Softmax (multinomial LR)	42	0.52 [0.39, 0.71]	48
Ridge classifier	26	0.32 [0.22, 0.48]	62
Linear SVM	15	0.19 [0.11, 0.31]	76
Gaussian naive Bayes	82	1.02 [0.83, 1.27]	16
Random forest	36	0.45 [0.32, 0.62]	52
LightGBM	45	0.56 [0.42, 0.75]	42
MLP (64 hidden units)	43	0.54 [0.40, 0.72]	46

Pooled counts treat repeated predictions of the same sample as separate trials, so the Wilson intervals are too narrow; descriptive only.

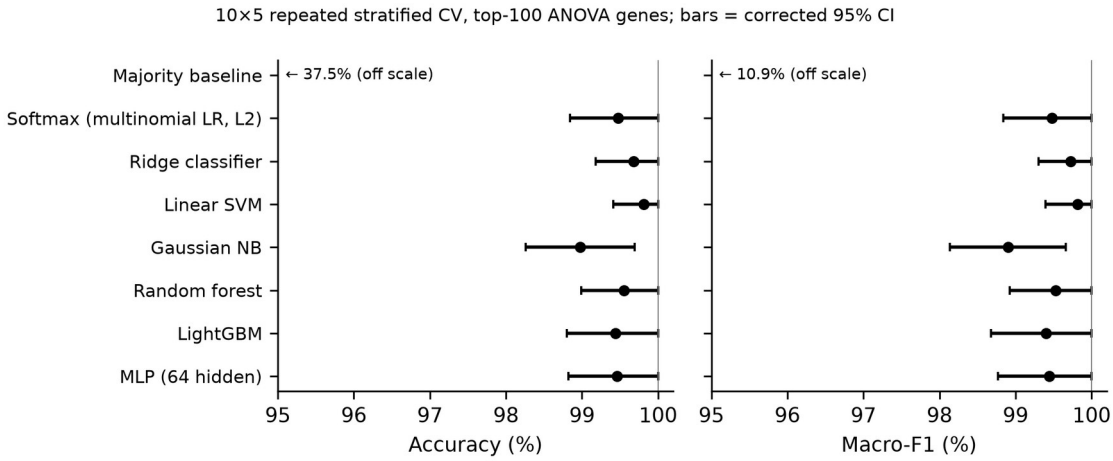


Figure S2. Accuracy and macro-F1 with corrected 95% confidence intervals. Mean over 50 folds of 10×5 repeated stratified CV; Nadeau–Bengio corrected 95% CI. Majority baseline off the axis.

S3 Feature-count ablation

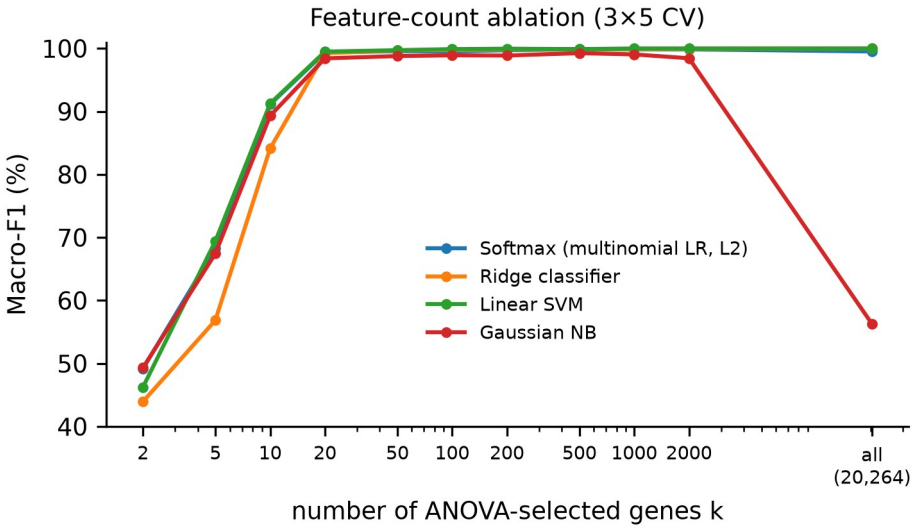


Figure S3. Macro-F1 as a function of the number of selected genes. Selection, scaling and classifier refitted inside every training fold; 3×5 repeated CV; log-scaled x-axis.

Table S4. Performance by number of selected genes, 3×5 repeated CV

Model	k	Accuracy, % (mean ± SD)	Macro-F1, % (mean ± SD)	Errors per repeat	Fit, ms
Softmax (multinomial LR)	2	63.80 ± 6.82	49.16 ± 6.42	290.0	416
Ridge classifier	2	61.22 ± 8.53	43.94 ± 8.52	310.7	367
Linear SVM	2	62.17 ± 8.02	46.24 ± 7.09	303.0	376
Gaussian naive Bayes	2	59.09 ± 10.26	49.32 ± 6.89	327.7	385
Softmax (multinomial LR)	5	75.86 ± 1.67	68.31 ± 3.41	193.3	412
Ridge classifier	5	72.24 ± 1.18	56.85 ± 2.17	222.3	387
Linear SVM	5	76.07 ± 2.05	69.43 ± 3.93	191.7	370
Gaussian naive Bayes	5	68.12 ± 3.75	67.43 ± 4.73	255.3	366
Softmax (multinomial LR)	10	92.67 ± 5.44	91.13 ± 6.30	58.7	393
Ridge classifier	10	89.22 ± 6.71	84.14 ± 9.66	86.3	397
Linear SVM	10	92.55 ± 5.63	91.27 ± 6.49	59.7	375
Gaussian naive Bayes	10	90.84 ± 7.67	89.32 ± 7.64	73.3	370
Softmax (multinomial LR)	20	99.46 ± 0.52	99.37 ± 0.60	4.3	399

Ridge classifier	20	99.29 ± 0.62	99.25 ± 0.72	5.7	382
Linear SVM	20	99.54 ± 0.55	99.50 ± 0.64	3.7	387
Gaussian naive Bayes	20	98.63 ± 0.86	98.43 ± 0.97	11.0	367
Softmax (multinomial LR)	50	99.62 ± 0.46	99.56 ± 0.57	3.0	542
Ridge classifier	50	99.63 ± 0.39	99.68 ± 0.34	3.0	384
Linear SVM	50	99.79 ± 0.45	99.72 ± 0.58	1.7	384
Gaussian naive Bayes	50	98.92 ± 0.90	98.79 ± 0.89	8.7	375
Softmax (multinomial LR)	100	99.50 ± 0.54	99.51 ± 0.54	4.0	425
Ridge classifier	100	99.71 ± 0.40	99.75 ± 0.34	2.3	398
Linear SVM	100	99.88 ± 0.26	99.89 ± 0.22	1.0	414
Gaussian naive Bayes	100	99.00 ± 0.78	98.91 ± 0.82	8.0	389
Softmax (multinomial LR)	200	99.75 ± 0.52	99.73 ± 0.58	2.0	414
Ridge classifier	200	99.83 ± 0.29	99.86 ± 0.24	1.3	391
Linear SVM	200	99.96 ± 0.16	99.96 ± 0.14	0.3	497
Gaussian naive Bayes	200	98.96 ± 0.61	98.88 ± 0.69	8.3	384
Softmax (multinomial LR)	500	99.83 ± 0.37	99.86 ± 0.32	1.3	481
Ridge classifier	500	99.88 ± 0.26	99.89 ± 0.22	1.0	432
Linear SVM	500	99.88 ± 0.26	99.90 ± 0.21	1.0	1333
Gaussian naive Bayes	500	99.46 ± 0.57	99.26 ± 0.80	4.3	409
Softmax (multinomial LR)	1000	99.83 ± 0.37	99.86 ± 0.32	1.3	555
Ridge classifier	1000	99.88 ± 0.26	99.89 ± 0.22	1.0	489
Linear SVM	1000	100.00 ± 0.00	100.00 ± 0.00	0.0	577
Gaussian naive Bayes	1000	99.33 ± 0.69	99.05 ± 0.91	5.3	389
Softmax (multinomial LR)	2000	99.83 ± 0.29	99.86 ± 0.24	1.3	1131
Ridge classifier	2000	99.96 ± 0.16	99.96 ± 0.14	0.3	563
Linear SVM	2000	100.00 ± 0.00	100.00 ± 0.00	0.0	752
Gaussian naive Bayes	2000	98.88 ± 0.72	98.45 ± 0.91	9.0	401
Softmax (multinomial LR)	all (≈20,264)	99.54 ± 0.55	99.52 ± 0.56	3.7	1435
Ridge classifier	all (≈20,264)	100.00 ± 0.00	100.00 ± 0.00	0.0	643
Linear SVM	all (≈20,264)	100.00 ± 0.00	100.00 ± 0.00	0.0	5979
Gaussian naive Bayes	all (≈20,264)	63.13 ± 3.41	56.28 ± 3.85	295.3	515

SD across 15 folds. 'all': every gene with non-zero variance in the training fold.

S4 All-gene benchmark

Table S5. All seven classifiers with every expressed gene, 10×5 repeated CV

Model	Accuracy, % [95% CI]	Macro-F1, % [95% CI]	Errors per repeat	Fit, s
Softmax (multinomial LR)	99.48 [98.91, 100.00]	99.45 [98.83, 100.00]	4.2	0.76
Ridge classifier	100.00 [100.00, 100.00]	100.00 [100.00, 100.00]	0.0	0.59
Linear SVM	100.00 [100.00, 100.00]	100.00 [100.00, 100.00]	0.0	6.81
Gaussian naive Bayes	63.10 [59.02, 67.18]	56.50 [51.44, 61.56]	295.6	0.55
Random forest	99.61 [99.14, 100.00]	99.68 [99.28, 100.00]	3.1	7.84
LightGBM	99.45 [98.81, 100.00]	99.40 [98.69, 100.00]	4.4	478.26
MLP (64 hidden units)	99.45 [98.65, 100.00]	99.36 [98.42, 100.00]	4.4	5.89

Zero-variance filter and z-scoring fitted in-fold, no ANOVA selection; same folds as the main benchmark. Fit times measured with parallel jobs; indicative only.

S5 All pairwise comparisons

Table S6. All 21 pairwise macro-F1 comparisons

A	B	Δ (B − A), pts	p	Holm p	TOST Holm p	P(A/ROPE/B)	MDD80
Linear SVM	Gaussian naive Bayes	-0.91	0.031	0.66	0.890	0.84/0.16/0.00	1.2
Ridge classifier	Gaussian naive Bayes	-0.83	0.063	1.00	0.890	0.77/0.23/0.00	1.2
Gaussian naive Bayes	Random forest	+0.63	0.147	1.00	0.890	0.01/0.38/0.62	1.2
Linear SVM	MLP	-0.37	0.176	1.00	0.110	0.32/0.68/0.00	0.8
Softmax	Gaussian naive Bayes	-0.58	0.203	1.00	0.890	0.57/0.42/0.01	1.3
Softmax	Linear SVM	+0.33	0.207	1.00	0.074	0.00/0.74/0.26	0.7
Gaussian naive Bayes	MLP	+0.54	0.217	1.00	0.890	0.01/0.45/0.54	1.2
Linear SVM	LightGBM	-0.41	0.232	1.00	0.328	0.40/0.59/0.01	1.0
Softmax	Ridge classifier	+0.24	0.247	1.00	0.006	0.00/0.89/0.11	0.6
Ridge classifier	MLP	-0.28	0.253	1.00	0.035	0.19/0.81/0.00	0.7
Ridge classifier	LightGBM	-0.33	0.323	1.00	0.178	0.30/0.69/0.01	0.9
Linear SVM	Random forest	-0.29	0.323	1.00	0.085	0.23/0.76/0.00	0.8
Gaussian naive Bayes	LightGBM	+0.50	0.327	1.00	0.890	0.03/0.47/0.50	1.4
Ridge classifier	Random forest	-0.20	0.408	1.00	0.013	0.11/0.89/0.00	0.7

Random forest	MLP	-0.08	0.638	1.00	< 0.001	0.01/0.99/0.00	0.5
Random forest	LightGBM	-0.13	0.661	1.00	0.028	0.10/0.88/0.02	0.8
Ridge classifier	Linear SVM	+0.09	0.680	1.00	< 0.001	0.00/0.97/0.03	0.6
Softmax	LightGBM	-0.08	0.801	1.00	0.038	0.10/0.86/0.04	0.9
Softmax	Random forest	+0.05	0.802	1.00	< 0.001	0.00/0.99/0.01	0.5
Softmax	MLP	-0.04	0.846	1.00	< 0.001	0.01/0.98/0.00	0.6
LightGBM	MLP	+0.04	0.893	1.00	0.028	0.05/0.88/0.08	0.9

Corrected resampled t-test (Nadeau–Bengio), Holm over 21 pairs; TOST margin ±1 point; Bayesian correlated t-test, ROPE ±0.5 point; MDD80 in points. Accuracy: results/pairwise_all.csv in File S1.

S6 Leakage audit

Table S7. Leaky vs in-fold selection, real labels, n = 801

k	Model	Accuracy, % leaky / in-fold (Δ, p)	Macro-F1, % leaky / in-fold (Δ, p)
10	Softmax	95.22 / 92.32 (+2.89, p = 0.27)	92.94 / 90.23 (+2.71, p = 0.40)
10	Linear SVM	95.37 / 92.31 (+3.06, p = 0.24)	93.26 / 90.65 (+2.61, p = 0.39)
10	Gaussian NB	95.47 / 90.80 (+4.66, p = 0.20)	93.52 / 88.98 (+4.54, p = 0.25)
20	Softmax	99.76 / 99.54 (+0.22, p = 0.39)	99.65 / 99.50 (+0.16, p = 0.60)
20	Linear SVM	99.78 / 99.55 (+0.22, p = 0.46)	99.68 / 99.50 (+0.19, p = 0.62)
20	Gaussian NB	99.00 / 98.71 (+0.29, p = 0.41)	98.74 / 98.53 (+0.20, p = 0.65)
100	Softmax	99.50 / 99.48 (+0.02, p = 0.87)	99.55 / 99.48 (+0.07, p = 0.73)
100	Linear SVM	99.78 / 99.81 (-0.04, p = 0.76)	99.79 / 99.81 (-0.02, p = 0.88)
100	Gaussian NB	98.98 / 98.98 (+0.00, p = 1.00)	98.90 / 98.90 (+0.00, p = 0.97)
1,000	Softmax	99.85 / 99.84 (+0.01, p = 0.79)	99.87 / 99.86 (+0.01, p = 0.79)
1,000	Linear SVM	100.00 / 100.00 (+0.00, p = 1.00)	100.00 / 100.00 (+0.00, p = 1.00)
1,000	Gaussian NB	99.21 / 99.20 (+0.01, p = 0.79)	98.95 / 98.94 (+0.02, p = 0.72)

10×5 repeated CV, identical folds; corrected paired t-test, unadjusted.

Permuted labels, n = 801 (softmax, 100 permutations, mean ± SD of 5-fold accuracy): k = 10: leaky 38.1% ± 1.2 (max 40.9), in-fold 33.0% ± 1.4 (max 36.3); k = 100: leaky 35.9% ± 2.3 (max 42.4), in-fold 25.3% ± 1.5 (max 29.1). Small samples (n = 60, 12 per class, 50 subsamples, softmax): permuted labels, k = 10: leaky 51.2% ± 7.0, in-fold 19.7% ± 5.4; permuted labels, k = 100: leaky 68.2% ± 5.8, in-fold 19.9% ± 6.3; real labels, k = 10: leaky 92.7% ± 8.2, in-fold 90.7% ± 5.8; real labels, k = 100: leaky 99.9% ± 0.4, in-fold 99.3% ± 1.1.

S7 Misclassified samples and quality metrics

Table S8. Samples misclassified by softmax in at least one repeat, with within-class quality z-scores

Sample	True	Predicted (count)	Repeats wrong	Mean expr. z	Zero frac. z	Centroid corr. z (rank)
sample_129	LUAD	BRCA ×9	9/10	-2.4	+1.4	-2.5 (5/141)
sample_507	LUAD	BRCA ×9	9/10	-1.9	+1.3	-2.1 (9/141)
sample_599	LUAD	BRCA ×9	9/10	-3.1	+2.1	-3.8 (1/141)
sample_510	COAD	LUAD ×8	8/10	+0.4	+0.1	-1.0 (10/78)
sample_61	KIRC	BRCA ×7	7/10	+1.2	-1.9	-3.3 (4/146)

z-scores within the true class. Centroid correlation: Pearson correlation with the mean profile of the other samples of the class; rank 1 = least typical.

Global summary statistics alone (9 per sample) predicted the class with 52.7% balanced accuracy in 5-fold CV (chance 20%).

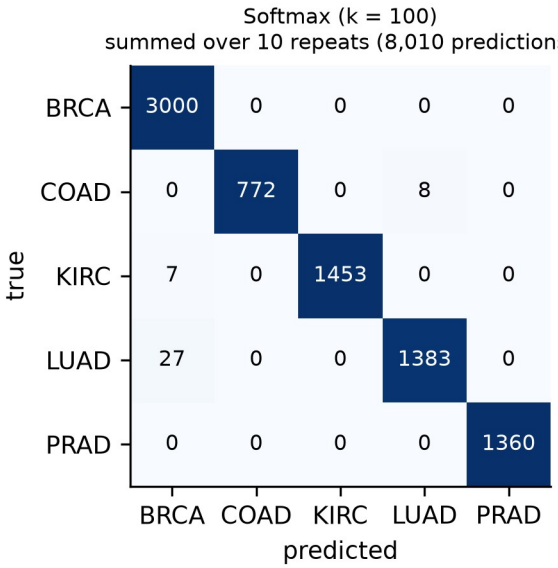


Figure S4. Confusion matrix of softmax regression. Counts summed over 10 repeats of 5-fold CV; colour: row-normalised recall.

S8 Top-ranked genes per class and selection stability

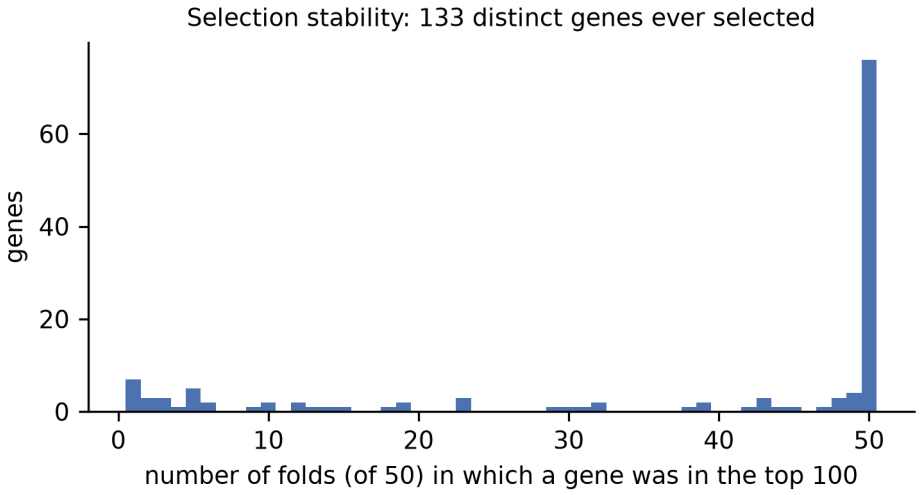


Figure S5. Stability of the top-100 ANOVA selection across folds. Number of training folds (of 50) in which each gene was among the top 100.

The genes below are the dummy names of the UCI release (gene_<index>, ordered as in the original Synapse submission syn4301332). Symbols are not distributed with the release, so the genes are not interpreted biologically.

Table S9. Five largest positive softmax coefficients per class

Class	Rank	Gene index	Coefficient	Selected in folds	Mean in class	Mean other
BRCA	1	gene_18746	0.450	50/50	12.97	8.17
BRCA	2	gene_5576	0.241	43/50	4.74	4.20
BRCA	3	gene_5577	0.198	48/50	4.04	3.58
BRCA	4	gene_17906	0.196	44/50	9.84	6.22
BRCA	5	gene_8348	0.190	50/50	9.76	6.78

COAD	1	gene_12013	0.322	50/50	11.35	1.78
COAD	2	gene_3523	0.263	50/50	11.82	2.13
COAD	3	gene_7238	0.250	50/50	11.79	2.46
COAD	4	gene_2037	0.221	50/50	6.86	0.29
COAD	5	gene_11464	0.199	39/50	11.23	3.84
KIRC	1	gene_3439	0.221	50/50	11.83	0.84
KIRC	2	gene_1858	0.218	50/50	10.01	0.71
KIRC	3	gene_6733	0.207	50/50	12.98	2.13
KIRC	4	gene_2129	0.161	49/50	10.51	3.23
KIRC	5	gene_13818	0.156	50/50	8.53	0.39
LUAD	1	gene_7964	0.416	50/50	8.64	5.10
LUAD	2	gene_15898	0.385	50/50	15.70	1.04
LUAD	3	gene_15895	0.375	50/50	10.45	0.23
LUAD	4	gene_11903	0.348	50/50	10.87	0.71
LUAD	5	gene_15894	0.327	50/50	10.49	1.06
PRAD	1	gene_13976	0.224	50/50	10.65	0.44
PRAD	2	gene_9175	0.197	50/50	16.75	1.12
PRAD	3	gene_9176	0.197	50/50	18.24	1.02
PRAD	4	gene_16358	0.194	50/50	14.94	7.42
PRAD	5	gene_18135	0.190	50/50	11.08	0.33

Softmax refitted on all samples (descriptive). Expression means on the released (log-scale) values.

Stability summary: 133 genes ever selected; 76 in all 50 folds; 85 in ≥ 90%; mean pairwise Jaccard 0.873 (min 0.786); 76 of the 100 genes selected on the full data were selected in every fold. Nogueira Φ = 0.931 (bootstrap 95% CI 0.929–0.937); average Kuncheva index 0.931.

S9 Computational footprint and fold-level distributions

Table S10. Computational footprint with 100 input genes (one CPU thread)

Model	Fit per fold incl. selection, ms	Prediction, ms/sample	Parameters or tree nodes	Model size, kB
Softmax (multinomial LR)	360	0.178	505	2.7
Ridge classifier	358	0.179	505	2.7
Linear SVM	374	0.175	505	4.5
Gaussian naive Bayes	360	0.176	1,005	4.5
Random forest	1037	0.296	11,910	1371.3
LightGBM	1690	0.221	21,444	1369.9
MLP (64 hidden units)	518	0.180	6,789	114.7

Fit: mean over 50 folds (640 training samples), including ANOVA on 20,531 genes. Size: pickled classifier only.

Shared virtual machine; indicative only.

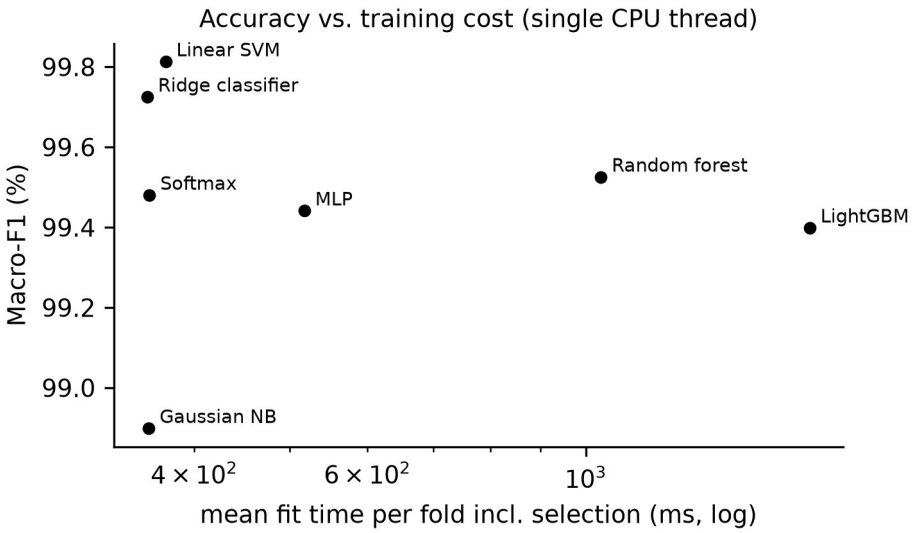


Figure S6. Macro-F1 against mean fit time per fold. Fit time includes zero-variance filtering, ANOVA on all genes and scaling (one CPU thread).

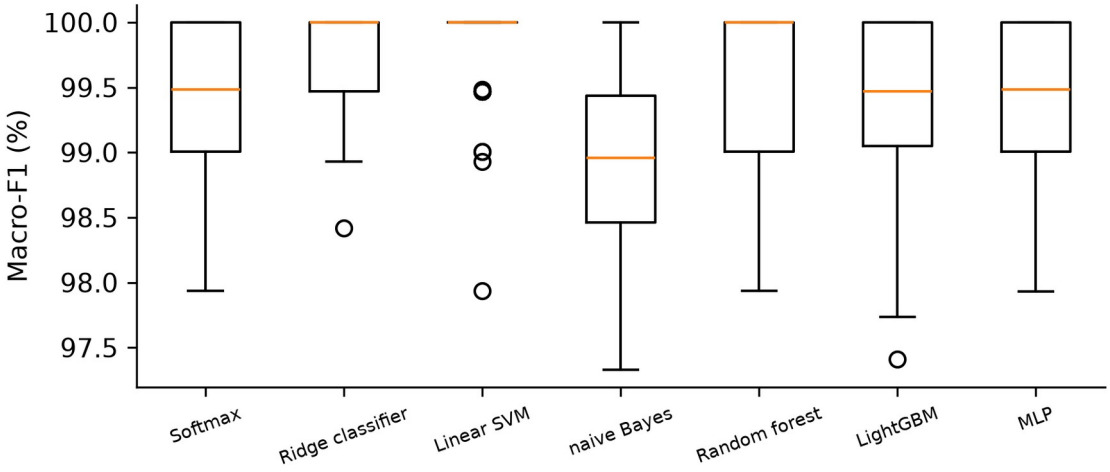


Figure S7. Distribution of fold-level macro-F1. 50 test folds per model (161 or 160 samples each).

S10 Reproducing the results

Supplementary File S1 (code zip) contains run_all.py, src/experiments.py, src/audit.py, src/novelty.py, src/robust.py, src/figures.py, requirements.txt and a README. Place data.csv and labels.csv from the UCI archive in data/ and run 'python run_all.py'; all tables in the paper and this supplement are generated from the files written to results/, and all figures to figures/ (300 dpi). 'python run_all.py --quick' runs a 2-repeat smoke test, '--revision-only' reruns only the revision analyses from saved fold results, '--revision-only --extension-only' reruns only the reliability and data-efficiency analyses, and '--jobs' sets the number of parallel workers.

S11 Calibration and conformal prediction, all models

Table S11. Calibration of pooled out-of-fold predictions (models fitted on whole training folds)

Model	Acc., %	Mean conf., %	ECE, % [95% CI]	Calibrated reference ECE, % [95% range]	Equal-mass ECE, %	Class-wise ECE, %	Log-loss	Brier
Softmax	99.48	99.57	0.47 [0.25, 0.83]	0.14 [0.08, 0.22]	0.22	0.25	0.0098	0.0058
Naive Bayes	98.98	99.99	1.02 [0.42, 1.69]	0.02 [0.01, 0.03]	1.01	0.41	0.3377	0.0203
Random forest	99.55	96.94	2.72 [2.32, 3.21]	0.40 [0.22, 0.61]	2.61	1.25	0.0372	0.0110
LightGBM	99.44	99.74	0.45 [0.17, 0.88]	0.11 [0.07, 0.17]	0.30	0.27	0.0237	0.0098
MLP	99.46	99.47	0.35 [0.24, 0.69]	0.17 [0.09, 0.26]	0.19	0.30	0.0135	0.0068

10×801 pooled predictions; ECE with 15 equal-width bins; CI: bootstrap over the 801 tumours (2,000 resamples). Reference: labels redrawn from the predicted probabilities (consistency resampling), which treats predictions as independent, so its range is somewhat too narrow. Class-wise ECE: mean over folds.

Softmax reliability bins (confidence range: predictions, errors): 0.33–0.40: 1, 1; 0.40–0.47: 3, 0; 0.47–0.53: 7, 2; 0.53–0.60: 16, 15; 0.60–0.67: 2, 2; 0.67–0.73: 10, 9; 0.73–0.80: 9, 5; 0.80–0.87: 13, 6; 0.87–0.93: 19, 2; 0.93–1.00: 7930, 0.

Table S12. Marginal conformal prediction sets: coverage spread, tie-breaking and cross-conformal variant

Model	Variant	Nominal, %	Coverage, %	Fold range, %	Beta 5th pct, %	Size	Singleton, %	Empty, %	Error on accepted, %	LUAD cov., %
Softmax	split	90	89.6	78.1–95.6	86.0	0.90	89.6	10.4	0.00	83.0
		95	94.7	85.0–99.4	92.0	0.95	94.7	5.3	0.00	89.9
		99	99.4	96.9–100.0	98.2	1.00	99.4	0.5	0.18	97.9
	split, randomised	90	89.5	78.1–95.6	86.0	0.90	89.5	10.5	0.00	82.7
		95	94.7	85.0–99.4	92.0	0.95	94.7	5.3	0.00	89.9
		99	98.9	94.4–100.0	98.2	0.99	99.0	0.9	0.18	96.8
	cross-conformal	90	90.7	82.0–95.0	86.0	0.91	90.7	9.3	0.00	85.3
		95	95.5	88.8–99.4	92.0	0.96	95.5	4.5	0.00	91.0
		99	99.1	96.9–100.0	98.2	0.99	99.2	0.8	0.03	96.9
Naive Bayes	split	90	98.6	96.2–100.0	86.0	1.00	99.8	0.2	1.28	99.7
		95	98.6	96.2–100.0	92.0	1.00	99.8	0.2	1.28	99.7
		99	99.9	97.5–100.0	98.2	4.52	12.0	0.0	0.94	100.0
	split, randomised	90	90.2	84.4–95.6	86.0	0.91	91.4	8.6	1.26	91.1
		95	94.5	87.5–97.5	92.0	0.96	95.7	4.3	1.30	95.9
		99	99.0	96.2–100.0	98.2	2.55	27.8	0.2	1.35	99.6
Random forest	split	90	90.1	82.6–95.6	86.0	0.90	90.1	9.9	0.00	82.9
		95	94.9	89.4–98.8	92.0	0.95	94.9	5.1	0.00	90.1
		99	99.3	96.2–100.0	98.2	1.00	99.1	0.5	0.15	98.0
	split, randomised	90	89.8	82.6–95.6	86.0	0.90	89.8	10.2	0.00	82.5
		95	94.8	89.4–98.1	92.0	0.95	94.8	5.2	0.00	90.1
		99	99.0	95.6–100.0	98.2	0.99	98.9	0.9	0.14	97.2
LightGBM	split	90	89.5	78.8–95.0	86.0	0.89	89.5	10.5	0.00	83.2
		95	95.0	87.5–99.4	92.0	0.95	95.0	5.0	0.00	91.3
		99	99.4	96.9–100.0	98.2	1.01	98.9	0.4	0.25	98.5
	split, randomised	90	89.4	78.8–95.0	86.0	0.89	89.4	10.6	0.00	83.2
		95	95.0	87.5–99.4	92.0	0.95	95.0	5.0	0.00	91.3
		99	99.0	96.3–100.0	98.2	1.00	99.0	0.7	0.28	97.9
MLP	split	90	90.0	80.1–96.9	86.0	0.90	90.0	10.0	0.00	84.4
		95	95.0	85.0–98.8	92.0	0.95	95.0	5.0	0.00	89.8

	99	99.5	96.9–100.0	98.2	1.00	99.2	0.4	0.16	98.5
split, randomised	90	89.8	80.1–96.9	86.0	0.90	89.8	10.2	0.00	84.2
	95	94.9	85.0–98.8	92.0	0.95	94.9	5.1	0.00	89.5
	99	98.9	91.9–100.0	98.2	0.99	98.9	0.9	0.23	96.9

Split: 25% of each outer training fold for calibration (n ≈ 161). Randomised: smoothed p-values breaking ties. Cross-conformal: 5 inner folds of the whole training fold (softmax only). Beta 5th pct: 5th percentile of coverage conditional on a calibration set of this size.

Table S13. Abstention rules for softmax at the 99% level: accepted predictions and errors

Rule	Accepted	Errors	Error rate, % [bootstrap 95% CI]	Wilson 95% CI
No abstention (75% model)	8,010	41	0.51 [0.42, 0.60]	[0.38, 0.69]
Split conformal, 99%	7,960	14	0.18 [0.10, 0.25]	[0.10, 0.30]
Max-probability, same threshold	7,973	21	0.26 [0.16, 0.35]	[0.17, 0.40]
Max-probability, abstention matched on calibration data	7,957	23	0.29 [0.16, 0.43]	[0.19, 0.43]
Cross-conformal, 99%	7,942	2	0.03 [0.00, 0.06]	[0.01, 0.09]
No abstention (cross-conformal ensemble)	8,010	50	0.62 [0.59, 0.66]	[0.47, 0.82]

Pooled over 10×801 test predictions. Bootstrap over the 10 repeats. Paired difference split conformal − no abstention: -0.34 points [-0.47, -0.21]. Wilson intervals treat repeated predictions as independent and are too narrow.

Table S14. Split-conformal prediction sets (no tie-breaking), marginal and Mondrian, all probabilistic models

Model	Sets	Nominal, %	Coverage, %	Size	Singleton, %	Empty, %	Acc. singletons, %	Coverage BRCA/COAD/KIRC/LUAD/PRAD, %
Softmax	marginal	90	89.60	0.90	89.6	10.4	100.00	88 / 90 / 95 / 83 / 93
	marginal	95	94.71	0.95	94.7	5.3	100.00	95 / 96 / 97 / 90 / 96
	marginal	99	99.36	1.00	99.4	0.5	99.82	100 / 99 / 99 / 98 / 100
	mondrian	90	91.15	0.91	91.1	8.9	100.00	90 / 95 / 89 / 92 / 93
	mondrian	95	96.28	1.87	13.4	0.0	74.45	95 / 100 / 97 / 96 / 97
	mondrian	99	100.00	5.00	0.0	0.0	—	100 / 100 / 100 / 100 / 100
Naive Bayes	marginal	90	98.56	1.00	99.8	0.2	98.72	98 / 97 / 99 / 100 / 100
	marginal	95	98.56	1.00	99.8	0.2	98.72	98 / 97 / 99 / 100 / 100
	marginal	99	99.89	4.52	12.0	0.0	99.06	100 / 100 / 100 / 100 / 100
	mondrian	90	98.74	1.38	61.9	0.1	98.82	98 / 98 / 99 / 100 / 100
	mondrian	95	99.10	2.33	5.0	0.0	98.80	98 / 100 / 99 / 100 / 100
	mondrian	99	100.00	5.00	0.0	0.0	—	100 / 100 / 100 / 100 / 100
Random forest	marginal	90	90.10	0.90	90.1	9.9	100.00	93 / 86 / 90 / 83 / 93
	marginal	95	94.87	0.95	94.9	5.1	100.00	97 / 95 / 93 / 90 / 97
	marginal	99	99.34	1.00	99.1	0.5	99.85	100 / 99 / 99 / 98 / 100
	mondrian	90	91.80	0.92	91.8	8.2	100.00	91 / 95 / 90 / 93 / 93
	mondrian	95	96.57	1.87	13.1	0.0	75.33	95 / 100 / 98 / 96 / 97
	mondrian	99	100.00	5.00	0.0	0.0	—	100 / 100 / 100 / 100 / 100
LightGBM	marginal	90	89.45	0.89	89.5	10.5	100.00	92 / 84 / 91 / 83 / 92
	marginal	95	94.98	0.95	95.0	5.0	100.00	96 / 93 / 96 / 91 / 97
	marginal	99	99.35	1.01	98.9	0.4	99.75	100 / 99 / 100 / 99 / 99
	mondrian	90	91.41	0.91	91.4	8.6	99.99	90 / 95 / 90 / 92 / 92
	mondrian	95	96.14	1.87	13.4	0.0	73.63	95 / 100 / 96 / 96 / 96
	mondrian	99	100.00	5.00	0.0	0.0	—	100 / 100 / 100 / 100 / 100

MLP	marginal	90	90.01	0.90	90.0	10.0	100.00	88 / 93 / 94 / 84 / 93
	marginal	95	95.01	0.95	95.0	5.0	100.00	96 / 97 / 96 / 90 / 97
	marginal	99	99.48	1.00	99.2	0.4	99.84	100 / 99 / 100 / 99 / 100
	mondrian	90	91.31	0.91	91.3	8.7	100.00	90 / 95 / 90 / 92 / 93
	mondrian	95	96.39	1.87	13.3	0.0	74.72	95 / 100 / 97 / 96 / 97
	mondrian	99	100.00	5.00	0.0	0.0	—	100 / 100 / 100 / 100 / 100

Score 1 − p(label). Naive Bayes over-covers at 90% and 95% because saturated probabilities produce tied scores (see randomised variant above). Mondrian sets need ≥ 19 (95%) or ≥ 99 (99%) calibration samples per class; available: about 60 BRCA, 29 KIRC, 28 LUAD, 27 PRAD, 16 COAD.

S12 Learning curves and leakage by sample size

Table S15. Learning curves (k = 100, in-fold), evaluated on full held-out folds

Model	Per class	Training n	Accuracy, % (± 95% CI)	Macro-F1, % (± 95% CI)	Fold SD of F1
Gaussian NB	3	15	70.22 ± 3.66	71.36 ± 3.34	10.74
Gaussian NB	5	25	66.18 ± 2.93	66.63 ± 3.23	11.07
Gaussian NB	10	50	74.02 ± 2.14	74.04 ± 2.36	6.95
Gaussian NB	20	100	88.99 ± 1.27	89.43 ± 1.05	3.98
Gaussian NB	40	200	96.40 ± 1.01	96.65 ± 0.96	2.00
Gaussian NB	60	300	97.72 ± 0.27	97.91 ± 0.26	1.36
Linear SVM	3	15	98.16 ± 0.80	98.32 ± 0.69	2.20
Linear SVM	5	25	98.94 ± 0.37	98.98 ± 0.34	1.38
Linear SVM	10	50	99.45 ± 0.22	99.45 ± 0.22	0.58
Linear SVM	20	100	99.58 ± 0.15	99.60 ± 0.15	0.57
Linear SVM	40	200	99.79 ± 0.07	99.82 ± 0.06	0.31
Linear SVM	60	300	99.80 ± 0.05	99.83 ± 0.04	0.31
Softmax	3	15	97.94 ± 0.74	98.12 ± 0.67	2.54
Softmax	5	25	98.91 ± 0.36	99.04 ± 0.29	1.17
Softmax	10	50	99.36 ± 0.20	99.40 ± 0.20	0.62
Softmax	20	100	99.48 ± 0.12	99.52 ± 0.12	0.54
Softmax	40	200	99.63 ± 0.06	99.66 ± 0.06	0.40
Softmax	60	300	99.58 ± 0.09	99.58 ± 0.07	0.49

Balanced random draws from each of the 50 outer training folds; 95% CI from the 10 repeat means (t interval).

Table S16. Leaky vs in-fold selection by dataset size (softmax accuracy, %)

Labels	k	n	Leaky	In-fold	Δ, points (± 95% CI)
permuted	10	30	73.3	19.6	+53.8 ± 3.6
permuted	10	60	52.6	19.1	+33.5 ± 3.4
permuted	10	120	43.1	19.3	+23.8 ± 2.4
permuted	10	240	35.3	19.7	+15.6 ± 1.4
permuted	10	390	30.7	20.5	+10.3 ± 1.4
permuted	100	30	85.8	20.7	+65.1 ± 4.8
permuted	100	60	68.9	19.7	+49.2 ± 4.7
permuted	100	120	55.2	19.6	+35.6 ± 3.5
permuted	100	240	41.0	20.0	+21.0 ± 1.8
permuted	100	390	34.7	20.6	+14.1 ± 1.7
real	10	30	88.6	85.4	+3.1 ± 2.5
real	10	60	94.3	90.6	+3.7 ± 2.0
real	10	120	94.2	93.6	+0.7 ± 1.3
real	10	240	91.1	91.6	-0.4 ± 1.5
real	10	390	89.7	91.3	-1.6 ± 0.8
real	100	30	100.0	99.0	+1.0 ± 0.7
real	100	60	100.0	99.2	+0.8 ± 0.5
real	100	120	99.8	99.6	+0.2 ± 0.1
real	100	240	99.7	99.5	+0.2 ± 0.1
real	100	390	99.7	99.5	+0.2 ± 0.1
permuted (all 801, unbalanced)	10	801	38.1	33.0	+5.1 ± 0.3
permuted (all 801, unbalanced)	100	801	35.9	25.3	+10.6 ± 0.5

Balanced subsamples: 30 per n, 5-fold CV each; CI over subsamples. At n = 390 every subsample contains all 78 COAD tumours, so subsamples overlap and the CIs are optimistic. Last rows: all 801 samples, 100 label permutations, a different design (chance 24.3% vs 20%).